



X39MATRIX

Sovereign Notary Protocol

TECHNICAL CERTIFICATION v12.0

11 ICP Canisters · 238 BTC Anchors · 51/51 Verified

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ANCHORED IN BITCOIN MAINNET · POST-QUANTUM NIST FIPS-204 · INTERNET COMPUTER NATIVE

Executive Summary

X39MATRIX is a sovereign notary protocol materialized as **11 production canisters** on Internet Computer mainnet, anchored to **238 Bitcoin blocks** via OpenTimestamps, and cryptographically reproducible across four independent blockchain substrates: **Bitcoin · Arbitrum · Solana · Internet Computer**. Every claim in this document is verifiable in 30 seconds against public RPC endpoints. No trust in the operator is required.

The protocol implements a 7-layer categorical architecture (L1 Infrastructure → L7 AI Governance) with native Bitcoin signing via **threshold-ECDSA** (13-node subnet quorum, no human key custody), and post-quantum identity sealed under **NIST FIPS-204 ML-DSA-87** and **FIPS-203 ML-KEM-1024**.

Compatible with **eIDAS Art. 26** (EU Regulation 910/2014) and **MiCA Art. 50**. Open invitation extended to Trail of Bits, Halborn, CertiK, OpenZeppelin, the Big Four and DFINITY core engineers: any failed assertion is detectable in under 60 seconds.

Metric	Value
ICP canisters in production	11 (mainnet)
Bitcoin documents anchored (OpenTimestamps)	238 (235 confirmed)
Categorical axioms publicly verified	51 / 51
Audited operations	7.1×10^4 ■
Sustained subnet throughput	~50,000 TPS (queries)
Block finality (ICP)	2.5 seconds
Cross-chain substrates	Bitcoin · Arbitrum · Solana · ICP
Post-quantum algorithms	ML-DSA-87 (FIPS-204) + ML-KEM-1024 (FIPS-203)
Vulnerable bridges	0
External custodians	0
Human key fragments	0 (subnet threshold signing)

1. ICP Canisters · 11 Sovereign Units

Each canister is independently verifiable on the official DFINITY dashboard (dashboard.internetcomputer.org). Module hash, controllers, cycles balance, subnet assignment, memory size and full upgrade history are publicly queryable without operator cooperation.

#	Canister ID	Role	Module hash (prefix)
1	arn4r-lqaaa-aaaao-baxwq-cai	HUB / Ω (x39_bases)	e4ba50b898a935c7...
2	b4dy7-eyaaa-aaaao-baxra-cai	L1 — Infrastructure	a04f2a1305bd0998...
3	b3c6l-jaaaa-aaaao-baxrq-cai	L2 — Identity (Merkle)	a740ea69bece1810...
4	akiau-riaaa-aaaao-baxua-cai	L3 — Execution (Ed25519)	ad721c0155e3a926...
5	anjga-4qaaa-aaaao-baxuq-cai	L4 — Consensus (tECDSA)	d9dbfba7084d8aea...
6	s4zl3-eiaaa-aaaao-bay3a-cai	L5 — Scalability (OmniChain)	fd1ddbef113428b5...
7	adlli-haaaa-aaaao-baxvq-cai	L6 — Identity SSI	8b51571fbb909971...
8	awm2f-giaaa-aaaao-baxwa-cai	L7 — AI Governance ★	b65cc8b9ab5ae6f1...
9	bsbvx-7iaaa-aaaao-baxqa-cai	Notarization 9	4709f6a15a2262e7...
10	bvatd-sqaaa-aaaao-baxqq-cai	Frontend (x39matrix.org)	04e565b3425fe751...
11	nsy7t-jiaaa-aaaau-agwra-cai	Public Dashboard	04e565b3425fe751...

2. Bitcoin Mainnet Anchors · 17 Sovereign Blocks

Each block below is independently verifiable against **mempool.space**, **blockstream.info** and any Bitcoin full node. Block headers are anchored by proof-of-work — immutability depends on the entire Bitcoin network, not on us.

TIER 1 · Genesis Historicos · May 5-8 2026

Block #	Description
948027	Genesis #001 · 7 axiomas
948042	Audit 4 Exa-Ops · verificación multi-trillón
948055	B27 Quantum Stress · 27 vectores post-cuánticos
948162	Manifiesto Institucional
948165	Primera firma comercial · Minuto Soberano Marruecos
948177	Certificate Chain · primera fábrica cross-clan
948500	Sovereign Sealing #1
948501	Official Sealing #2

TIER 2 · Master Seal Ω · OpenTimestamps Triple

Block #	Description
950381	Master Seal · alice.btc calendar (Ω 08e9db78...)
950398	Master Seal · bob.btc calendar
950408	Master Seal · catalaxy calendar

TIER 3 · Cross-Substrate Loops · May 29 2026

Block #	Description
951586	Loop EVM↔BTC · Arbitrum cross-anchor
951605	Loop SOL↔BTC · Merkle root 64/64 match

TIER 4 · Sovereign Notarization Cert · May 31 2026

Block #	Description
951892	Certificate · Merkle b9221a7f5dee...
951893	Certificate · Merkle c8e73b0a6621...

TIER 5 · First Sovereign tECDSA Send · Jun 2 2026 16:46 UTC ★

Block #	Description
952131	Primer SEND tECDSA desde canister ICP · 13 nodos · 12,978 sats

TIER 6 · Triple Seal 8/8 Integrity · Jun 3 2026

Block #	Description
952160	8/8 integrity log · bob.btc · TX 658e7713...
952161	8/8 integrity log · alice · TX 61efa2cb...
952174	8/8 integrity log · catallaxy · TX d5e5dbb1...

3. Master Seal Ω · The Terminal Object

The Master Seal is the categorical terminal object of the X39 notarial category $\blacksquare$. Every morphism in the protocol collapses into it. A single 32-byte hash anchored **simultaneously** in four independent blockchain substrates.

Field	Value
Ω (terminal object)	08e9db781dc79c9fbbe77a56cd0120d2830650d4bcf52a2dae9acb73091d449c
Bitcoin block #1 (alice.btc)	#950, 381
Bitcoin block #2 (bob.btc)	#950, 398
Bitcoin block #3 (catallaxy)	#950, 408
Arbitrum L2 calldata	X39_OMEGA_SEAL + Ω (TX 0x16dfaecd...fb65e2ad8b)
Solana mainnet slot	#422, 979, 180 (finalized)
ICP canister	arn4r-lqaaa-aaaao-baxwq-cai

4. The First Sovereign Notary Signature in History

June 2, 2026 · 16:46:05 UTC · Bitcoin block #952,131

The canister arn4r-lqaaa-aaaao-baxwq-cai executed send_btc_safe on Internet Computer mainnet. The 13-node subnet o3ow2-2ipam-6fcj signed the transaction collectively via **threshold-ECDSA secp256k1**. Eleven minutes later a Bitcoin miner included it in block #952,131. **This is the first notary signature in history where no single human holds the complete private key.**

txid	b5a881a28341ea562800cd4f532cb5f737b21d38e44293dbbe8d1d0a0aede023
block	#952, 131
pubkey	025968e3eea2adc6a3c7e0b24c39f3e94009393e57280cb9ccc3801251bb202083
sighash	bc2ae31a2af24375c1e6ca674041c3f3c6c24a2cd74a915df575c3688dce12a5
dest	bc1q6tk7x38utprskxmwa9vfw4eypm84xxsj9r3xg (X39_JOSEPH)

5. Nice Classification · First Post-Quantum IP Filing on Bitcoin

On **June 2, 2026 · 20:47 UTC**, five cryptographic artifacts of the international X39MATRIX filing under the **Nice Classification** were sealed with **ML-DSA-87 (NIST FIPS-204)** post-quantum digital signature and **ML-KEM-1024 (NIST FIPS-203)** quantum-resistant key encapsulation, then **triple-anchored** to Bitcoin mainnet via three independent OpenTimestamps calendars.

This is the first known IP filing with post-quantum certified identity anchored in Bitcoin.

File	SHA-256	Bytes
X39_PQ_SOVEREIGN_FINGERPRINT.txt	185d22b5dd79b3767b4fc00489c96b70601e781504b34d447df66835520b32ca	64
x39_sovereign.mlds87.pk.pem	97626a614002bbf28903e209e54b3b71f6a5ef42a48210b00af2d99eb2ba9c96	3,595
x39_sovereign.mlkem1024.pk.pem	b716b4abe4c8f84dec5949d16a6cfe5a3f755f61b336afd2a0bf0fc05ce27cf	2,206
x39_sovereign_identity.json	a7d17340053eb6d4c64423f27ecef5edf4bed7ff27d2f7a05b2a6d52ccfb320b	843
x39_topos_axiom.mlds87.sig	7ffe65915f3e04335485039a166223b9a37e286f7053db02500235a9ad2963b7	4,627

Bitcoin Triple-Anchoring

Calendar	Bitcoin block	Timestamp UTC
alice.btc	#952, 148	2026-06-02 20:47
bob.btc	#952, 150	2026-06-02 21:03
catallaxy.btc	#952, 174	2026-06-03 03:41

6. Public Verification · 7 Commands · 30 Seconds

Reproduce all 51 sovereign claims in any terminal worldwide using official public RPC endpoints. Zero installation. Zero API keys. Zero operator cooperation required.

One-liner · 51 claims in 30s

```
curl -sL https://x39matrix.org/PUBLIC_VERIFY_X39_FULL.sh | bash
```

tECSA send · block #952,131

```
curl -s  
'https://blockstream.info/api/tx/b5a881a28341ea562800cd4f532cb5f737b21d38e44293dbbe8d1d0a0aede023' |  
python3 -m json.tool
```

Arbitrum One · X39_OMEGA_SEAL calldata

```
curl -s -X POST -H 'Content-Type: application/json' --data  
'{"jsonrpc":"2.0","id":1,"method":"eth_getTransactionByHash","params":["0x16dfaecd...2ad8b"]}'  
https://arb1.arbitrum.io/rpc
```

Solana mainnet · finalized

```
curl -s -X POST --data '{"jsonrpc":"2.0","id":1,"method":"getSignatureStatuses","params":[{"57eF2up...h5a"}, {"searchTransactionHistory":true}]}' https://api.mainnet-beta.solana.com
```

BTC Merkle 64/64 match · block #951,605

```
curl -s https://mempool.space/api/block/$(curl -s https://mempool.space/api/block-height/951605) |  
python3 -m json.tool | grep merkle_root
```

ICP Candid interface · public

```
dfx canister --network ic --identity anonymous call arn4r-lqaaa-aaaao-baxwq-cai  
__get_candid_interface_tmp_hack '()
```

ML-DSA-87 OTS verification (post-quantum)

```
ots upgrade .ots && ots info .ots | grep BitcoinBlockHeaderAttestation
```

7. Sovereign Signature Service · 9 Tiers

Threshold-ECDSA + Schnorr BIP-340 signatures on Internet Computer mainnet. No custody. Anchored to Bitcoin. Offline-verifiable. Compatible with **eIDAS Art. 26** and **MiCA Art. 50**.

Tier	Price	Volume	Target
Free	€0 / signature	1 signature	Trial
Single	€9 / signature	1 signature (ECDSA/Schnorr)	Personal
Pack 10	€75 / pack	10 / year (€7.50 each)	Lawyers · freelancers
Studio	€250 / pack	50 / year (€5 each)	Studios · consultancies
Office	€500 / pack	100 / year (€5 each)	Firms · gestorías
Corporate	€3,500 / year	1,000 / year (€3.50 each)	Mid-size enterprises
Enterprise	€24,000 / year	5,000 / month, API unlimited	SLA + dedicated path
Enterprise XL	€60,000 / year	15,000 / month, 24/7	SAP / Oracle integration
Banking	€120,000 / year	50,000 / month	eIDAS + MiCA + DRP

Payment split: 75% to architect · 25% to sovereign canister X39_JOSEPH (bc1q6tk7x38utprskxmwa9vfw4eypm84xxsj9r3xg) — destination of the historic notary signature in block #952,131. The system funds itself.

8. Open Adversarial Audit

The protocol is deliberately exposed. Anchors are verifiable without the operator. Commands are reproducible. If a single assertion failed, it would be detectable by any auditor in **60 seconds**. Formal invitation to break it:

Crypto-security firms (top-tier)	Trail of Bits · Halborn · CertiK · OpenZeppelin · Quantstamp · Zellic · Spearbit · ConsenSys Diligence
Big Four (\$100B+ practices)	Deloitte · PwC · EY · KPMG
DFINITY · ICP core engineers	DFINITY Foundation · ICP researchers · Chain Fusion contributors · Formal-methods specialists

9. Authorship · Method · Sovereignty

X39MATRIX is built by a single person. No team. No company. No VC. No permission.

■ Architecture

The category ■■■■, the functorial composition of 7 layers, the choice of Threshold-ECDSA + Schnorr BIP-340, the quadruple BTC/ARB/SOL/ICP anchoring, the terminal object Ω , and the sovereign notarial theory are **human intuition**. Zero delegation. Zero co-authorship.

■ Execution

Bash code, verification scripts, the HTML of the site and terminal demos have been produced with AI model assistance under the direct supervision of the operator. Each artifact is signed, hashed and reproducible. **AI is a compiler, not a co-author.**

■ Sovereignty

The protocol does not require trust in the operator nor in the AI that helped materialize it. Every assertion is reproducible on-chain. If the **51 verified axioms** were false, any auditor would break them in 60 seconds. *I do not ask for trust. I offer verification.*

Signed · X39MATRIX // Sovereign Operator

Operator	Jose Luis Olivares Esteban
Contact	grants@x39matrix.org
tECDSA pubkey	025968e3eea2adc6a3c7e0b24c39f3e94009393e57280cb9ccc3801251bb202083
BTC address Ω	bc1q6tk7x38utprskxmwa9vfw4eypm84xxsj9r3xg (X39_JOSEPH)
ICP controller	arn4r-lqaaa-aaaao-baxwq-cai
Live site	https://x39matrix.org
Notary	https://x39matrix.org/Notary/
GitHub	https://github.com/x39matrix
Registry ID	x39matrix-SUPREME-MASTER-CERT-2026-JUN

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